

Technology Stack

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being - HDI Predictor
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, Jinja2</i>	<i>Simple and responsive frontend for Flask web application with prediction form</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python 3.x, Flask</i>	<i>Lightweight Python web framework used for ML model deployment and routing</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV File - Kaggle HDI Dataset</i>	<i>Stores HDI data for 195 countries - lightweight and easy to process with Pandas</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Local - Flask Development Server</i>	<i>Local deployment at http://127.0.0.1:5000 - easy setup without cloud costs</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Version control for project code submission and team collaboration</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle</i>	<i>ML model training (Linear Regression), data processing, visualization and model serialization</i>